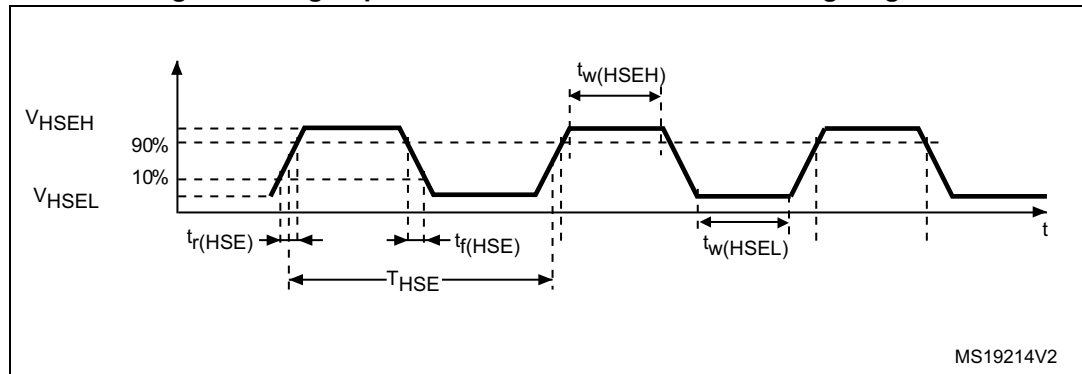


1. Specified by design. Not tested in production.

Figure 15. High-speed external clock source AC timing diagram



Low-speed external user clock generated from an external source

In bypass mode the LSE oscillator is switched off and the input pin is a standard GPIO.

The external clock signal has to respect the I/O characteristics in [Section 5.3.13](#). See [Figure 16](#) for recommended clock input waveform.

Table 38. Low-speed external user clock characteristics

Symbol	Parameter ⁽¹⁾	Conditions	Min	Typ	Max	Unit
f_{LSE_ext}	User external clock source frequency	-	-	32.768	1000	kHz
V_{LSEH}	OSC32_IN input pin high level voltage	-	$0.7 \times V_{DDIO1}$	-	V_{DDIO1}	V
V_{LSEL}	OSC32_IN input pin low level voltage	-	V_{SS}	-	$0.3 \times V_{DDIO1}$	
$t_{w(LSEH)}/t_{w(LSEL)}$	OSC32_IN high or low time	-	250	-	-	ns

1. Specified by design. Not tested in production.

Figure 16. Low-speed external clock source AC timing diagram

